

Supplementary Material

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S1 The Swiss Sector-Coupled Energy System

This section describes the Swiss sector-coupled energy system model. Section S1.1 discusses the demands for each sector, Section S1.2 provides the techno-economic parameters for all conversion and storage technologies, Section S1.3 shows the calculation of GWP100-based climate change impacts, and Sections S1.4 and S1.5 provide more details on electricity and heat-generating technologies.

S1.1 Demands

Sections S1.1.1, S1.1.2, and S1.1.3 describe how demands are calculated for non-industrial electricity, residential heat, and private transportation, respectively. Regarding spatial resolution, the demands are aggregated for all of Switzerland, which is represented as a 1-node system. Regarding temporal resolution, the demands are resolved hourly for the year 2050. In many cases, we start with data for the year 2019 and project it for the year 2050 based on assumptions from literature. Industrial energy demands are discussed in Section 2.2 of the main paper.

S1.1.1 Non-industrial electricity demand

The hourly electricity demand profile is derived from data provided by SwissGrid (Swissgrid AG, 2020), who provide electricity consumption in 15-minute increments for the year 2019. As we consider the year 2050, we perform several adaptations to the 2019 electricity demand: First, as the Swiss population is projected to grow from 8.69 million inhabitants in 2019 to 10.44 million inhabitants in 2050 (Federal Statistical Office (FSO), 2020), we scale the electricity demand accordingly. Second, we assume a 20% electricity demand reduction in 2050 due to efficiency gains (The Swiss Confederation, Fedlex - The publication platform for the federal law, 2021; Swiss Federal Office of Energy (SFOE), 2013; Swiss Federal Office of Energy (SFOE), Prognos AG, 2012). We exclude the 6 TWh yearly industrial electricity demand since these are accounted for in the modelling of industrial production (Section 2.2 of the main paper).

Additionally, we exclude the electricity demands used for residential heating applications, since these demands are accounted for in the heating sector (Section S1.1.2). Heat demands for space heating and warm water are provided by the Swiss Federal office of Energy (SFOE)(Swiss Federal Office of Energy

(SFOE), Prognos AG, TEP Energy GmbH, INFRAS AG, 2020). The demands are grouped by the energy sources used to supply the demands. In 2019, 13% of the heat demand of private households and 5.5% of the heat demand of the commercial sector was provided by electricity for electric boilers and heat pumps. To resolve double counting, we subtract the power consumption for residential heat from the electricity demand.

Our assumptions result in a final electricity demand of 45.2 TWh for 2050.

S1.1.2 Residential heat demand

Our model's residential heat demand is composed of the space heating and warm water demands of private households and the commercial sector, provided by the Swiss Federal office of Energy (SFOE) (Swiss Federal Office of Energy (SFOE), Prognos AG, TEP Energy GmbH, INFRAS AG, 2020). The national residential heat demand for Switzerland in 2019 was 74.7 TWh. As we consider the year 2050 in our model, we apply a 11% reduction to the heat demand (Scenario ZERO in Swiss Federal Office of Energy (SFOE) (2021)). This heat demand reduction accounts for climate change and the projected population growth.

To represent the residential heat demand in time, we use the approach of Baumgärtner et al. (2021). The residential heat demands for the commercial sector are assumed constant throughout the year. Following Baumgärtner et al. (2021), we model the temporal profile of the private household residential heat demand using the hourly outdoor temperature of Switzerland (Renewables.ninja, 2021; Gelaro and et al., 2017) as follows:

- 1) Days are assumed to have a heating demand only if the average daily outside temperature is above 12°C (Swiss Federal Office of Energy (SFOE), 2020b). Otherwise, the heat demand of the entire day is assumed zero.
- 2) For days that have a heat demand, the hourly demand is proportional to the normalized temperature difference between the target indoor temperature (20°C (Swiss Federal Office of Energy (SFOE), 2020b)) and the outdoor temperature. This temperature difference is normalized by the sum of the temperature difference for the entire year. Outdoor temperatures are taken from Renewables.ninja (2021); Gelaro and et al. (2017) for the year 2019.

S1.1.3 Private transportation demand

For the total private transportation demand in 2050, we use the forecast from the STEM model (Panos and Ramachandran, 2017; Prognos AG, 2012) of 67.1×10^9 vehicle kilometers (vkm). We derive an hourly demand profile f_t^v based on traffic census data (Federal Statistical Office (FSO), 2014) that includes average motorized vehicle counts per hour of a day at different measure points in Switzerland. We use data from ten measure points from the last traffic census in 2013 (Figure S1).

To derive the hourly private transportation demand profile for our model f_t^v , we normalize the hourly traffic counts of each measure point by the average and take the mean of the ten measure points at each hour. The resulting mean profile represents the ten measure points well (Figure S2).

The hourly private transportation demand is calculated with (Eq. S1). Every day in our model follows the same profile. Note further that any motorized vehicle is counted in the traffic census and not only private transportation. However, we assume that our hourly profile is proportional to passenger car traffic and is thus applicable to the private transportation demand.

$$V_t^{pc} = f_t^v \cdot V^{pc} \cdot \frac{1}{8760 h}, \forall t \in \mathcal{T}. \quad (\text{S1})$$

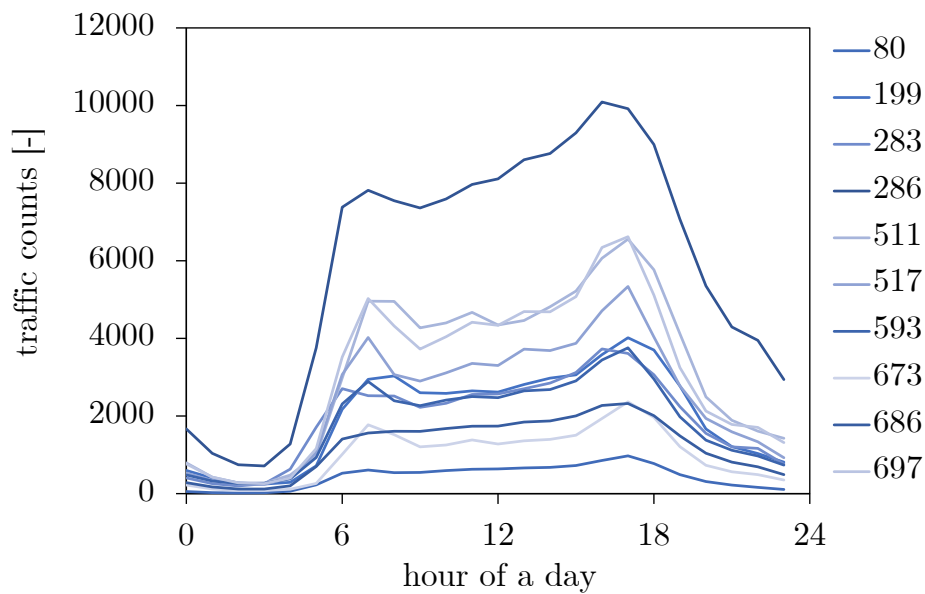


Figure S1 Daily traffic counts at ten measure points in Switzerland (80-697, category: ‘DTV Richt. 1+2’) in 2013 (Federal Statistical Office (FSO), 2014). The traffic counts describe how many motorized vehicles have on average been counted at each hour of the day.

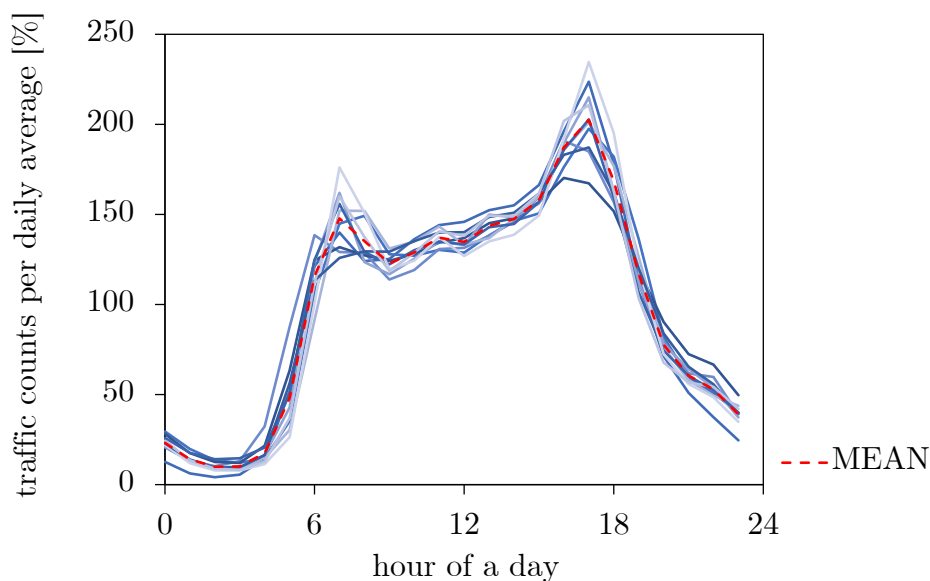


Figure S2 Normalized daily traffic profiles at ten selected measure points (compare to Figure S1). The red line represents the mean used for our hourly demand profile.

S1.2 Techno-economic parameters of conversion and storage technologies

The characteristics of all technologies included in our model are listed and described below. Details about electricity and heat sector technologies that require additional input data are found in Sections S1.4 and S1.5. In this section, we first present the costs and lifetimes (Section S1.2.1), including a more detailed explanation of the direct air capture (DAC) costs. We then present the technology efficiencies and capacity factors (Section S1.2.2).

S1.2.1 Costs and lifetimes

In our model, all costs are given in EUR, implemented with a fixed average exchange rate between EUR and CHF of 2019 (1:1.1125) (Federal Tax Administration (FTA), 2020). Except for direct air capture, which has operating costs, all technologies in our model have investment and fixed, yearly maintenance costs. The operational costs consist purely of fuel costs and are thus already counted in the import costs.

We annualize the investment costs of technologies with an interest rate of 5% and we use the minimum of the technology lifetime and the economic investment time of 30 years for the annualization, as described in Baumgärtner et al. (2021). The lifetimes of all technologies are listed in Table S1, S2, S3, and S4. Following the Swiss Energy strategy (Swiss Federal Office of Energy (SFOE), 2021), we assume 100% electrification in private transportation in 2050. Thus, we assume that all transportation demand is satisfied by battery electric vehicles (BEVs). As we focus on energy provision, we do not include the investment costs of electric vehicles in our cost calculation.

Table S1. Costs and lifetimes of electricity and heat production technologies for the year 2050 used in our Swiss energy system model.

Technology	CAPEX [EUR/kW]	Maintenance [EUR/(kW·a)]	Cost sources	Lifetime [years]
biogas-to-power	8,568	483.60	Bauer et al. (2017, 2019)	30 (Deutsche Energie-Agentur GmbH (dena), 2018)
electrode boiler civil	238	4.75	Brauner (2019)	20 (Henning and Sauer, 2015)
electrode boiler LT	238	4.75	Brauner (2019)	20 (Henning and Sauer, 2015)
electrode boiler MT	238	4.75	Brauner (2019)	20 (Henning and Sauer, 2015)
thermal insulation	2,369	0.00	Baumgärtner et al. (2021)	50 (Baumgärtner et al., 2021)
geothermal	10,301	417.98	Moret (2017b,a)	30 (Deutsche Energie-Agentur GmbH (dena), 2018)
heat pump (ground) civil	472	20.21	Moret (2017b,a)	20 (Henning and Sauer, 2015)
heat pump (ground) LT	472	20.21	Moret (2017b,a)	20 (Henning and Sauer, 2015)
hydrogen-to-power	575	9.20	Palzer (2016)	25 (Palzer, 2016)
large dam /PSH hybrid	3,146	44.94	Bauer et al. (2017, 2020)	100 (Deutsche Energie-Agentur GmbH (dena), 2018)
large dam pure	3,146	44.94	Bauer et al. (2017, 2020)	100 (Deutsche Energie-Agentur GmbH (dena), 2018)
gas boiler MT	57	1.13	Moret (2017b,a)	20 (Baumgärtner et al., 2021)
gas boiler HT	57	1.13	Moret (2017b,a)	20 (Baumgärtner et al., 2021)
gas combined cycle	845	20.67	Bauer et al. (2017, 2020)	20 (Baumgärtner et al., 2021)
photovoltaics	817	16.98	Bauer et al. (2017, 2020)	25 (Deutsche Energie-Agentur GmbH (dena), 2018)
power-to-hydrogen	300	5.00	Deutsche Energie-Agentur GmbH (dena) (2018)	16 (Deutsche Energie-Agentur GmbH (dena), 2018)
power-to-methane	591	14.40	Deutsche Energie-Agentur GmbH (dena) (2018)	20 (Deutsche Energie-Agentur GmbH (dena), 2018)
run-of-river	3,146	44.94	Bauer et al. (2017, 2020)	100 (Deutsche Energie-Agentur GmbH (dena), 2018)
onshore wind	2,023	67.42	Bauer et al. (2017, 2019)	25 (Deutsche Energie-Agentur GmbH (dena), 2018)

Table S2. Costs and lifetimes of storage technologies for the year 2050 used in our Swiss energy system model.

Technology	CAPEX [EUR/kWh]	Maintenance [EUR/kWh]	OPEX [EUR/kWh]	Cost sources	Lifetime [years]
Li-ion battery	350	12.00	-	Deutsche Energie-Agentur GmbH (dena) (2018)	20 (Deutsche Energie-Agentur GmbH (dena), 2018)
hydrogen lined rock cavern	1.8	-	0.026	Papadakis and Ahluwalia (2021)	30 (Papadakis and Ahluwalia, 2021)
gas storage tank	2.06	-	-	Schmidt et al. (2019) ^a	18 (Schmidt et al., 2019)
pumped storage hydro (pure)	3,146	44.94	-	Bauer et al. (2017, 2020)	100 (Deutsche Energie-Agentur GmbH (dena), 2018)

^a converted from hydrogen storage cost using densities at 25°C and 100bar

Table S3. Costs and lifetimes of transportation sector technologies for the year 2050 used in our Swiss energy system model.

Technology	CAPEX [EUR/car]	Maintenance [EUR/(car·year)]	Cost sources	Lifetime [years]
battery electric vehicles (BEV)	Not considered ^a	724.50	Sacchi et al. (2020b,a)	14.08 (Baumgärtner et al., 2021)

^a Details in Section S1.2.1

Table S4. Costs and lifetimes of direct air capture (DAC) for the year 2050 used in our Swiss energy system model.

Technology	CAPEX [EUR/(ton CO ₂ /h)]	OPEX [EUR/ton CO ₂]	Cost sources	Lifetime [years]
DAC	1,592,000	58.33	Fasihi et al. (2019); Metz et al. (2005)	30 (Deutz and Bardow, 2021; Fasihi et al., 2019)

Direct Air Capture (DAC)

We model a temperature-vacuum swing adsorption process for DAC. This DAC process consumes electricity and low temperature heat in order to capture CO₂ (Deutz and Bardow, 2021). The costs for DAC plants are displayed in Table S4. The operating costs include 1) the CO₂ capture process, 2) the CO₂ transport, and 3) the CO₂ storage in the North Sea. Each operating cost component is explained below. We use an exchange rate of 1.1196 USD/EUR (Federal Tax Administration (FTA), 2020).

- 1) The CO₂ capture process operating cost corresponds to 4% of the CAPEX, taken from the year 2020 as a conservative assumption (Fasihi et al., 2019).
- 2) The captured CO₂ must be transported from Switzerland to the nearest carbon storage site, which is Northern Lights (Furre et al., 2019) in the North Sea. We calculate the transport costs using an onshore pipeline from Switzerland to Rotterdam, where the CO₂ is loaded onto a ship and delivered to the storage site. The Intergovernmental Panel on Climate Change (IPCC) provides costs for pipeline transport with a mass flow rate of 6 Mt of CO₂ per year (Metz et al., 2005). We take an average distance of 800 km (reference: Lucerne-Rotterdam), given as 7.144 EUR (8 USD) per transported ton of CO₂. We reduce the costs for pipeline transport by 30% due to expected scale effects by 2050 (Jakobsen

et al., 2016). Ship transport to the Northern Lights storage site is expected to cost 5.36 EUR/ton CO₂ in 2050 (Metz et al., 2005). Total cost of transport is thus 10.36 EUR/ton CO₂.

3) The costs for permanent storage are 12.95 EUR/ton CO₂ (Metz et al., 2005).

S1.2.2 Efficiencies and secured capacity factors

As explained by Baumgärtner et al. (2021), time series aggregation removes extreme load points from the optimization. Therefore, installed capacities of generation and storage technologies are likely to be underestimated. The secured capacity factor is introduced as a parameter to guarantee that the total available production through generation and storage of a given product is greater than the peak product demand, comprising both exogenous and sector-coupled endogenous demands.

Table S5. Efficiencies and secured capacity factors of technologies used in our Swiss energy system model for the year 2050. Unless specified, all values are taken from Baumgärtner et al. (2021). Where no efficiency is given, the efficiency is implicitly included in the availability timeseries of the technologies.

Technology	Efficiency ^a or COP ^b	Secured Capacity Factor
biogas-to-power	/	88%
electrode boiler civil	99%	100%
electrode boiler LT	99%	100%
electrode boiler MT	99%	100%
thermal insulation	/	100%
geothermal	/	90%
heat pump (ground) (Deutz and Bardow, 2021)	2.51	100%
heat pump (ground) LT (Deutz and Bardow, 2021)	2.51	100%
hydrogen-to-power	26.7 $\frac{\text{MWh}_{\text{elec}}}{\text{ton}}$	100%
large dam hybrid	/	40%
large dam pure	/	40%
gas boiler MT	95%	100%
gas boiler HT	95%	100%
gas combined cycle	60%	86%
photovoltaics	/	1%
power-to-hydrogen	0.024 $\frac{\text{ton}}{\text{MWh}_{\text{elec}}}$	100%
power-to-methane	58.5%	100%
run-of-river	/	40%
onshore wind	/	5%
DAC ^c	500 MWh _{elec} 1500 MWh _{LTheat}	100%

^a heat generated per energy source consumed

^b coefficient of performance

^c for the capture of 1000 ton CO₂

Table S6. Efficiencies, secured capacity factors, and energy-to-power ratios of storage technologies for the year 2050 used in our Swiss energy system model. Unless specified, all values are taken from Baumgärtner et al. (2021).

Technology	Round-trip Efficiency	Secured Capacity Factor	Energy-to-Power ratio [hrs]
lithium-ion battery	93%	90%	1.25
pumped storage hydro (hybrid)	80% (Swiss Federal Office of Energy (SFOE), 2020a)	90%	2252 (Sec. S1.4.1)
pumped storage hydro (pure)	80% (Swiss Federal Office of Energy (SFOE), 2020a)	90%	2252 (Sec. S1.4.1)
hydrogen lined rock cavern	96.3% (Berger et al., 2020)	100%	400 (Lord et al., 2014)
gas storage tank	99% (Berger et al., 2020)	100%	1 ^a

^a high-end assumption for complete discharge in 1hr

Table S7. Average mileage and power consumption of transport technologies for the year 2050 used in our Swiss energy system model. The values are taken from Bundesministerium für Verkehr und digitale Infrastruktur (2020).

Technology	Average mileage [vehicle km]	Power consumption [kWh/100km]
battery electric vehicle (BEV)	13,600 (Bundesministerium für Verkehr und digitale Infrastruktur, 2020)	14.0 (Baumgärtner et al., 2021)

S1.3 Life Cycle Inventories

Life cycle inventories (LCIs) are used calculate the operational climate change impacts (GWP100), which we constrain to net-zero. We use the same inventories as Baumgärtner et al. (2021) while updating the data to ecoinvent 3.7.1 APOS (Wernet et al., 2016) (rather than 3.5 APOS) and changing the locations from Germany to Switzerland wherever possible.

Processes in the LCIs can contain energy-related environmental impacts from, for instance, electricity production. However, since we explicitly model the energy sectors in our energy system, the energy-related environmental impacts are already accounted for within our system boundary. To prevent double counting, we delete the energy-related environmental impacts that are already accounted for in the Swiss energy sectors. For the deletion, we follow the approach of Volkart et al. (2018) and Reinert et al. (2021). Moreover, as our model considers the year 2050, we employ the dynamic LCA method from Reinert et al. (2021) with the updated ecoinvent 3.7.1.

S1.4 Renewable electricity technologies

The sections below describe the modelling details beyond techno-economic parameters of electricity generation technologies that require more details. The techno-economic parameters, including costs, lifetimes, efficiencies, secured capacity factors, and lifetimes, for all technologies included in the model can be found in Section S1.2. Unless otherwise specified, the data for electricity generation technologies is taken from the open-source platform Open Power Systems Data (OPSD), which is divided into data for conventional (Open Power System Data, 2020a) and renewable (Open Power System Data, 2020b) power plants. Additionally, unless otherwise stated, techno-economic parameters are taken from Baumgärtner et al. (2021).

S1.4.1 Hydropower

Hydropower includes electricity production and storage technologies and well as combinations of the two. In our model, we implement four different hydropower technologies: run-of-river, pure large dam, pure pumped storage hydro (PSH), and a hybrid hydropower plant consisting of a large dam with PSH. Run-of-river power plants use the energy from flowing water to produce electricity. Large dams accumulate flowing water in a reservoir, subsequently releasing it for electricity production. PSH stores energy by pumping water into a reservoir at a higher elevation, producing electricity from the previously pumped water. Hybrid hydropower plants consist of both a PSH and a large dam, producing electricity from both natural inflow and previously pumped water (Figure S3).

To model the hybrid hydropower plant, we divide the hybrid large dam and hybrid PSH into two separate technologies and couple them via the power production capacity, P^h [MW], equating it to both the power production capacity for hybrid large dam, $P^{ld,h}$, and to the power production/storage capacity for the hybrid PSH, $P^{psh,h}$ (Equation S2). The power production/storage capacity for the hybrid PSH is calculated by multiplying the hybrid PSH energy storage capacity, $E^{psh,h}$ [MWh], by its flow-to-storage capacity factor, f^{psh} . To calculate the flow-to-storage capacity factor, we divide the total existing Swiss PSH turbine capacity, $P^{psh,tc}$ (3.52 GW (Open Power System Data, 2020a)), by 90% of the total existing storage volume of Swiss PSH reservoirs, $E^{psh,sc}$ (8.8 TWh (Felix et al., 2020)) (Equation S3). The 90% is assumed due to the reported 10% of unused reservoir volume that is always kept as backup (Swiss Federal Office of Energy (SFOE), 2020a; Felix et al., 2020).

To limit the simultaneous production of the hybrid large dam, $P_t^{ld,h}$, and the hybrid PSH system, $P_t^{psh,h}$, at a given time step, t , we add a constraint that limits the total simultaneous production from the hybrid system to its existing power production capacity (Equation S4).

$$P^h = P^{ld,h} = P^{psh,h} = E^{psh,h} \cdot f^{psh} \quad (\text{S2})$$

$$f^{psh} = \frac{P^{psh,tc}}{0.9 \cdot E^{psh,sc}} = 4.4 \cdot 10^{-4} \quad (\text{S3})$$

$$P_t^{ld,h} + |P_t^{psh,h}| \leq f^{psh} \cdot E^{psh,h} \quad \forall t \in \mathcal{T} \quad (\text{S4})$$

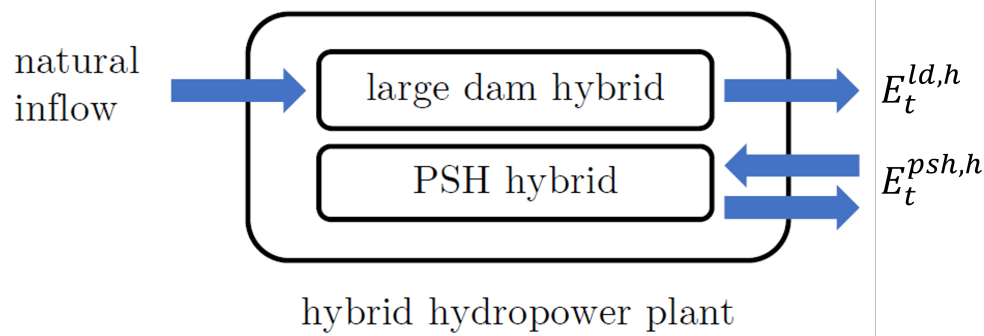


Figure S3 Energy flows of a hybrid hydropower plant consisting of a large dam and a pumped storage hydro (PSH) system. At a given time step, t , electricity can either be generated through natural inflow via the large dam, $E_t^{ld,h}$, or stored or withdrawn via the PSH system, $E_t^{psh,h}$.

The existing and potential capacities for the 4 hydropower technologies in 2050 are given in Table S8. Data for existing capacities for all hydropower technologies in 2019 are taken from a combination of the OPSD data (Open Power System Data, 2020a) and from the official Swiss statistics on hydropower plants (Swiss Federal Office of Energy (SFOE), 2018). According to a study by the Swiss Federal Office of Energy (SFOE) (Swiss Federal Office of Energy (SFOE), 2019c), the Swiss hydropower production is expected to increase by 1.56 TWh/a until 2050. The study distinguishes between expected increase for large hydropower plants ($> 10\text{MW}$), which we allocate to large dams (both pure and hybrid), and for small hydropower plants ($< 10\text{MW}$), which we allocate to run-of-river.

The hourly electricity generation from run-of-river and large dams is limited to the natural inflow of water. We calculate the electricity availability time series based on reported monthly electricity generation levels by hydropower technology (Swiss Federal Office of Energy (SFOE), 2019a). We set the monthly time series as the ratio of the monthly production and the maximum potential production based on the existing capacities. We assume the same availability for every hour in a given month.

Table S8. Existing and potential capacities for hydropower technologies in 2050. Data for existing capacities is taken from Open Power System Data (2020a); Swiss Federal Office of Energy (SFOE) (2018) and data for potential capacities from Swiss Federal Office of Energy (SFOE) (2019c). Cost and lifetime parameters are given in Tables S1 and S2. Efficiencies are given in Tables S5 and S6.

Technology	Existing Capacity	Potential Capacity
run-of-river	4.45 GW	4.71 GW
pure large dam	8.27 GW	8.49 GW
pure pumped storage hydro (PSH) ^a	1.2 TWh/528 MW	1.2 TWh/528 MW
hybrid large dam with PSH ^a	6.7 TWh/2.98 GW	6.8 TWh/3.04 GW

^a Capacities for PSH consider a flow-to-storage capacity factor of $4.4 \times 10^{-4} \frac{\text{MWh}}{\text{MW}}$.

S1.4.2 Photovoltaics and Wind Power

To derive the hourly production profiles of photovoltaic (PV) and wind technologies, both the maximum potential capacity and the hourly capacity factors are needed. Capacity factors refer to the share of the

maximum potential capacity that can be generated depending on the temporal availability of renewables. Our maximum potential PV capacity for the year 2050 is 67 TWh (Federal Department of the Environment, Transport, Energy and Communications (DETEC), 2018; Swiss Federal Office of Energy (SFOE), 2019b). We assume the scenario where PV covers roofs and facades of houses. Our potential wind capacity for the year 2050 is 4.3 TWh (Swiss Federal Office of Energy (SFOE), 2021). Our hourly capacity factors are derived from Renewables.ninja (Renewables.ninja, 2021; Pfenninger and Staffell, 2016; Staffell and Pfenninger, 2016), who provides a history of hourly capacity factors for PV and wind power based on weather data.

S1.4.3 Biogas and Geothermal

Our biogas-to-power electricity capacity includes all biomass and biogas power plants, sewage and landfill gas power plants, and other bioenergy and renewable waste plants from the OPSD data for Switzerland, yielding a total generation capacity of 87.2 MW (Open Power System Data, 2020b). All biogas technologies are represented by a single technology with a constant hourly availability of 57% (Kannan and Turton, 2011), or 50 MWh. We justify this simplified implementation since bioenergy technologies are similar and have a low overall contribution to the Swiss electricity mix now and in future decades (Swiss Federal Office of Energy (SFOE), 2013). Biogas-to-power plants are modeled as a pure power-producing technology, without heat cogeneration.

Regarding geothermal electricity, we set the potential capacity in 2050 to 2 TWh (Swiss Federal Office of Energy (SFOE), 2021) and assume a constant hourly availability.

S1.5 Heating technologies

S1.5.1 Heat pumps and electrode boilers

Heat pumps, electrode boilers, and gas burners are used to supply heat at different temperature levels (Table 2). However, heat pumps are not an option for the entire low temperature (LT) range in our model. Similarly, electrode boilers are not an option over the entire medium temperature (MT) range. Although heat pumps with a heat supply for temperatures above 100°C are already under development, they are not commercially available (Arpagaus et al., 2017). Similar holds true for electrode boilers, for which different sources define a maximum temperature level between 350°C-500°C (Antoni et al., 2015; Forschungsstelle für Energiewirtschaft e.V. (FfE), 2017; PARAT Halvorsen AS, 2021). Therefore, we limit the potential capacities of LT heat pumps and MT electrode boilers in our model. The LT heat pump's capacities are constrained to supply at maximum the industrial process heat demand between 0-100°C, and the MT electrode boiler's capacities are constrained to the demand between 200°C-400°C. However, LT heat pumps in our model are free to supply additional heat for DAC.

S1.5.2 Energetic rehabilitation

We constrain energetic rehabilitation such that it can at maximum provide 50% of the civil heat demand. This implementation is in accordance with a study by Palzer on Germany (Palzer, 2016), who show that the rehabilitation increases the thermal insulation and results in a 50% reduction of the heating demand of a building. The Swiss advisory platform Energieheld (Energieheld Schweiz, 2021) also refers to German data (co2online, 2021) when defining the reduction potential for Swiss buildings to be around 50%.

S2 Additional Figures

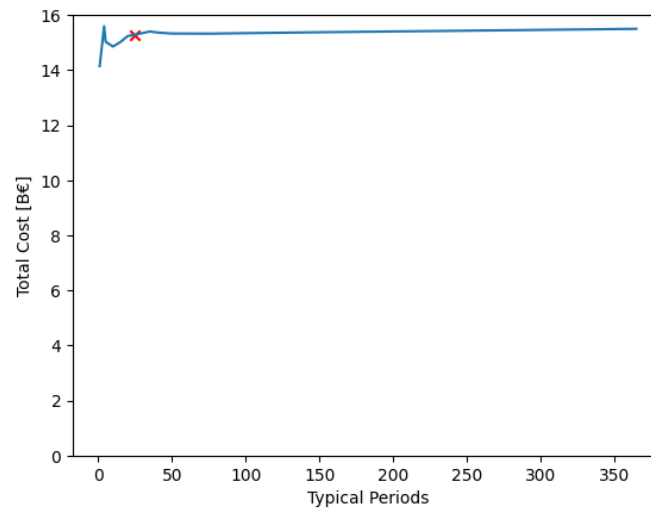


Figure S4 Total cost of the optimized energy system, in billion euro, depending on the number of typical days used for the time series aggregation. Red cross corresponds to our selected 25 typical days.

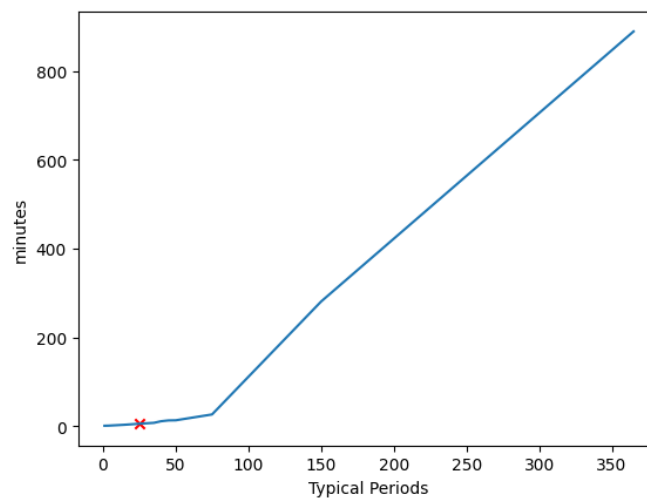


Figure S5 Solution time of the energy system optimization depending on the number of typical days used for the time series aggregation. Red cross corresponds to our selected 25 typical days.

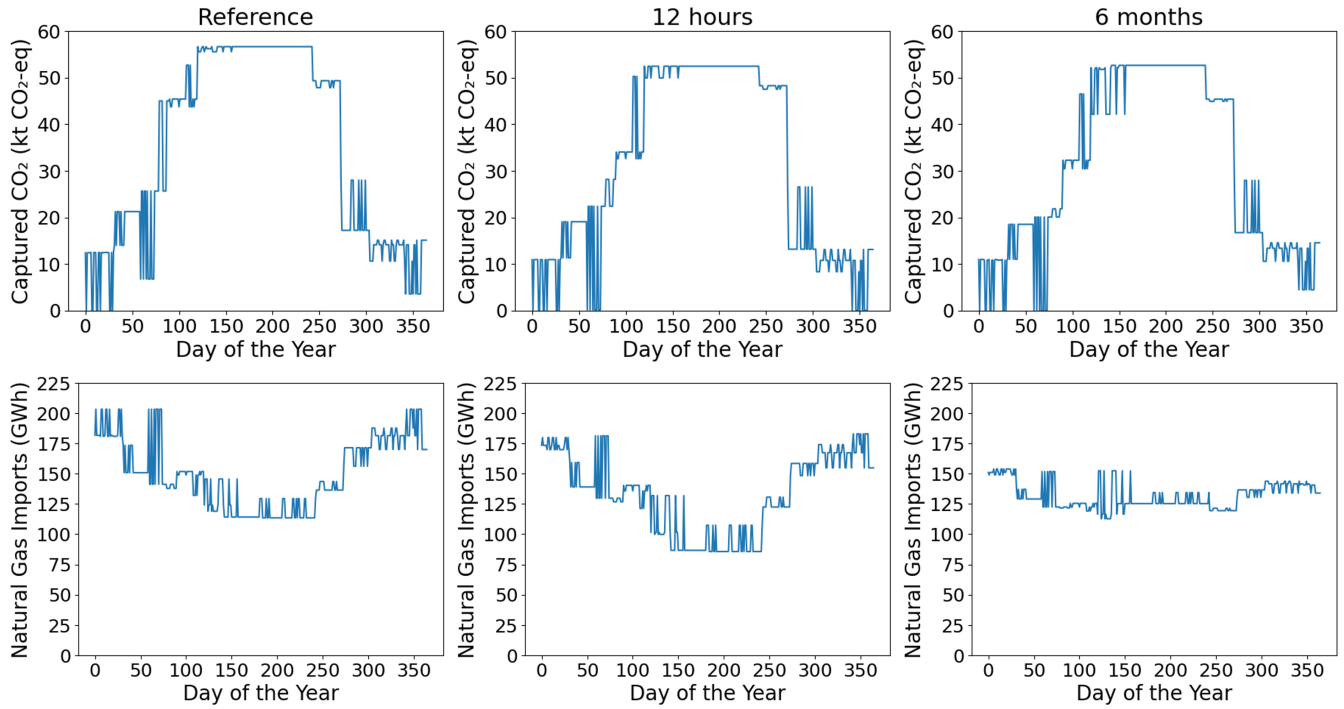


Figure S6 *New Low-Cost Dispatch* scenario: yearly profiles for CO₂ captured (top row) and natural gas imports (bottom row) for varying storage capacities (columns). The Reference case corresponds to no industrial DSM ($iflex = 0$). The 12-hour and 6-month storage capacity plots assume 100% load-shifting potential ($iflex = 1$) and 100% discharge efficiency ($\eta = 1$).

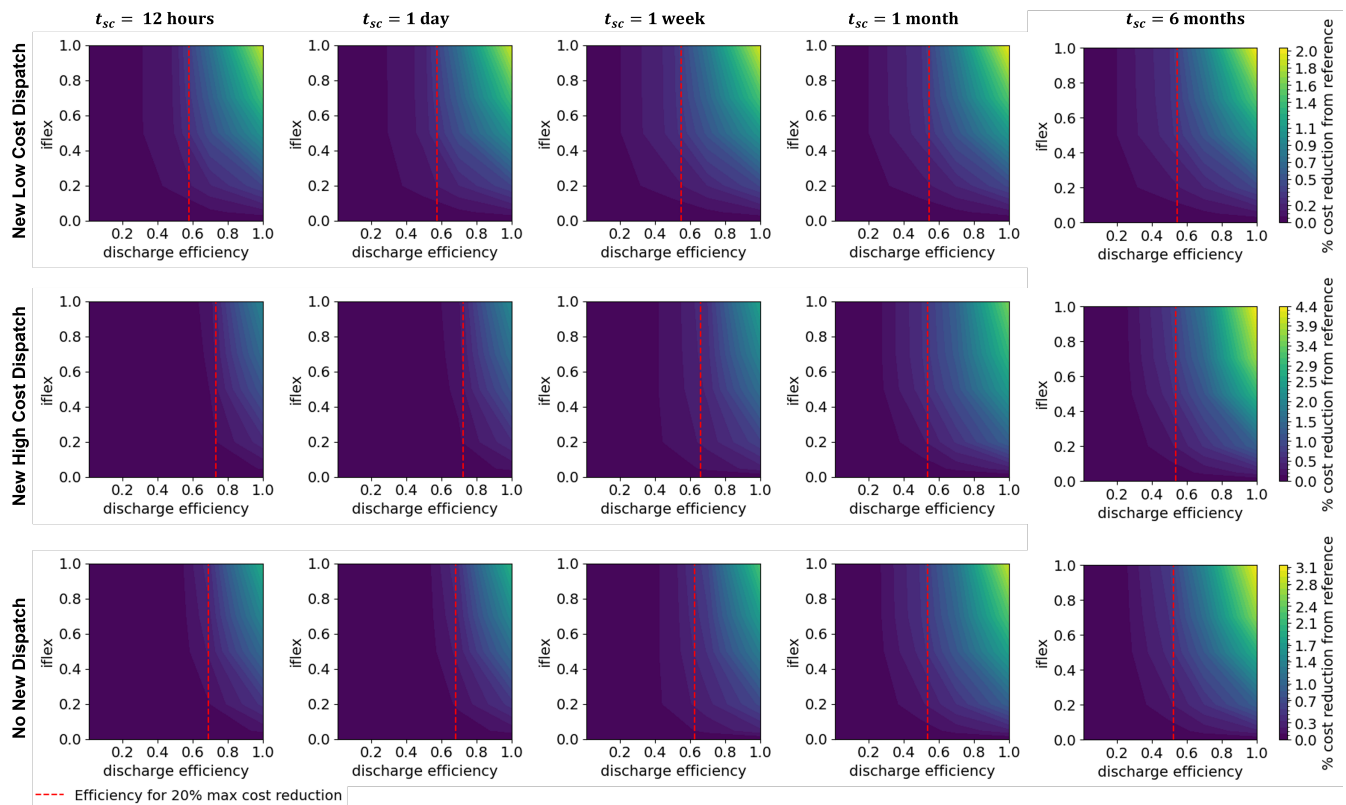


Figure S7 % cost reduction for the net-zero sector-coupled system for different values of load-varying potential, $iflex$, and discharge efficiency, η , vs. the reference case of no industrial DSM. Rows correspond to scenarios and columns correspond to the storage capacity in terms of the storage capacity time interval, t_{sc} , over which the demand for goods, d_{goods}^{base} , can accumulate. Red lines indicate the efficiency corresponding to 20% of the maximum cost reduction for the respective storage size, when considering $iflex = 1$.

References

- [Dataset] Antoni, O., Sager, C., and Gerhardt, N. (2015). Wärme-Roadmap und Handlungsempfehlungen. Fraunhofer-Institut für Energiewirtschaft und Energiesystemtechnik (IEE)
- [Dataset] Arpagaus, C., Bless, F., Schiffmann, J., and Bertsch, S. (2017). Review on High Temperature Heat Pumps – Market Overview and Research Status. doi:10.13140/RG.2.2.32913.94564
- Bauer, C., Hirschberg, S., Bäuerle, Y., Biollaz, S., Calbry-Muzyka, A., Cox, B., et al. (2017). *Potentials, costs and environmental assessment of electricity generation technologies*. Tech. rep., PSI, WSL, ETHZ, EPFL
- [Dataset] Bauer, C., Hirschberg, S., Bäuerle, Y., Biollaz, S., Calbry-Muzyka, A., Cox, B., et al. (2019). Potentials, costs and environmental assessment of electricity generation technologies, dataset version 2019-03-29
- [Dataset] Bauer, C., Hirschberg, S., Bäuerle, Y., Biollaz, S., Calbry-Muzyka, A., Cox, B., et al. (2020). Potentials, costs and environmental assessment of electricity generation technologies, dataset version 2020-03-05
- Baumgärtner, N., Deutz, S., Reinert, C., Nolzen, N., Kuepper, L. E., Hennen, M., et al. (2021). Life-Cycle Assessment of Sector-Coupled National Energy Systems: Environmental Impacts of Electricity, Heat, and Transportation in Germany Till 2050. *Frontiers in Energy Research* 9, 27. doi:10.3389/fenrg.2021.621502
- Berger, M., Radu, D., Fonteneau, R., Deschuyteneer, T., Detienne, G., and Ernst, D. (2020). The role of power-to-gas and carbon capture technologies in cross-sector decarbonisation strategies. *Electric Power Systems Research* 180, 106039. doi:https://doi.org/10.1016/j.epsr.2019.106039
- Brauner, G. (2019). *Systemeffizienz bei regenerativer Stromerzeugung* (Springer Vieweg, Wiesbaden), 1 edn. doi:10.1007/978-3-658-24854-3
- Bundesministerium für Verkehr und digitale Infrastruktur (2020). *Verkehr in Zahlen 2020/2021*. Tech. rep.
- [Dataset] co2online (2021). Accessed: June 19, 2021
- [Dataset] Deutsche Energie-Agentur GmbH (dena) (2018). dena-Leitstudie Integrierte Energiewende - Impulse für die Gestaltung des Energiesystems bis 2050
- Deutz, S. and Bardow, A. (2021). Life-cycle assessment of an industrial direct air capture process based on temperature–vacuum swing adsorption. *Nature Energy* 6, 203–213. doi:10.1038/s41560-020-00771-9
- [Dataset] Energieheld Schweiz (2021). Dämmung in der Schweiz - so können Sie sparen. Accessed: June 19, 2021
- Fasihi, M., Efimova, O., and Breyer, C. (2019). Techno-economic assessment of CO₂ direct air capture plants. *Journal of Cleaner Production* 224, 957–980. doi:https://doi.org/10.1016/j.jclepro.2019.03.086
- [Dataset] Federal Department of the Environment, Transport, Energy and Communications (DETEC) (2018). Solarenergiepotenziale der Schweizer Gemeinden - Version 01.01.2021
- [Dataset] Federal Statistical Office (FSO) (2014). DE/FRAutomatische Strassenverkehrszählung
- Federal Statistical Office (FSO) (2020). DE/FRSzenarien zur Bevölkerungsentwicklung der Schweiz und der Kantone 2020-2050 , 84
- [Dataset] Federal Tax Administration (FTA) (2020). Jahresmittelkurse 2019
- Felix, D., Müller-Hagmann, M., and Boes, R. (2020). Ausbaupotenzial der bestehenden Speicherseen in der Schweiz. *Wasser Energie Luft* 1/2020, 1–10
- [Dataset] Forschungsstelle für Energiewirtschaft e.V. (FfE) (2017). Kurzstudie Power-to-X
- Furre, A.-K., Meneguolo, R., Ringrose, P., and Kassold, S. (2019). Building confidence in CCS: From Sleipner to the Northern Lights Project. *First Break* 37, 81–87. doi:10.3997/1365-2397.n0038

- Gelaro, R. and et al., W. M. (2017). The Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA-2). *Journal of Climate* 30, 5419 – 5454. doi:10.1175/JCLI-D-16-0758.1
- Henning, H.-M. and Sauer, D. U. (2015). *Demand-Side-Management im Wärmemarkt - Technologiesteckbrief zur Analyse "Flexibilitätskonzepte für die Stromversorgung 2050"*. Tech. rep.
- Jakobsen, J., Roussanaly, S., and Anantharaman, R. (2016). A techno-economic case study of CO₂ capture, transport and storage chain from a cement plant in Norway. *Journal of Cleaner Production* 144, 523–539. doi:10.1016/j.jclepro.2016.12.120
- Kannan, R. and Turton, H. (2011). *Documentation on the Development of the Swiss TIMES Electricity Model (STEM-E)*. Tech. rep., Paul Scherrer Institut (PSI)
- Lord, A. S., Kobos, P. H., and Borns, D. J. (2014). Geologic storage of hydrogen: Scaling up to meet city transportation demands. *International Journal of Hydrogen Energy* 39, 15570–15582. doi:https://doi.org/10.1016/j.ijhydene.2014.07.121
- Metz, B., Davidson, O., de Coninck, H., Loos, M., and Meyer, L. (2005). *IPCC Special Report on Carbon Dioxide Capture and Storage*. Tech. rep., Intergovernmental Panel on Climate Change (IPCC)
- [Dataset] Moret, S. (2017a). Energy conversion technologies in SES - version 2017-12-12
- Moret, S. (2017b). *Strategic planning under uncertainty*. Ph.D. thesis, École polytechnique fédérale de Lausanne (EPFL). doi:10.5075/epfl-thesis-7961
- [Dataset] Open Power System Data (2020a). Data Package Conventional power plants - Version 2020-10-01. doi:10.25832/conventional_power_plants/2020-10-01
- [Dataset] Open Power System Data (2020b). Data Package Renewable power plants - Version 2020-05-20. doi:10.25832/renewable_power_plants/2020-08-25
- Palzer, A. (2016). *Sektorübergreifende Modellierung und Optimierung eines zukünftigen deutschen Energiesystems unter Berücksichtigung von Energieeffizienzmaßnahmen im Gebäudesektor*. Tech. rep., Fraunhofer-Institut für Solare Energiesysteme ISE
- [Dataset] Panos, E. and Ramachandran, K. (2017). Transport demand in stem, version 2017-12-15
- Papadias, D. and Ahluwalia, R. (2021). Bulk storage of hydrogen. *International Journal of Hydrogen Energy* 46, 34527–34541. doi:https://doi.org/10.1016/j.ijhydene.2021.08.028
- [Dataset] PARAT Halvorsen AS (2021). PARAT Elektrodenkessel Referenzen. Accessed: June 28, 2021
- Pfenninger, S. and Staffell, I. (2016). Long-term patterns of European PV output using 30 years of validated hourly reanalysis and satellite data. *Energy* 114, 1251–1265. doi:10.1016/j.energy.2016.08.060
- Prognos AG (2012). *Die Energieperspektiven für die Schweiz bis 2050*. Tech. rep.
- Reinert, C., Deutz, S., Minten, H., Dörpinghaus, L., von Pfingsten, S., Baumgärtner, N., et al. (2021). Environmental impacts of the future German energy system from integrated energy systems optimization and dynamic life cycle assessment. *Computers & Chemical Engineering* 153, 107406. doi:10.1016/j.compchemeng.2021.107406
- [Dataset] Renewables.ninja (2021). Renewables.ninja. Accessed: April 22, 2021
- [Dataset] Sacchi, R., Bauer, C., Cox, B., and Mutel, C. (2020a). Car technicoeconomic data and performance projections - version 2020-04-27
- Sacchi, R., Bauer, C., Cox, B., and Mutel, C. (2020b). carculator: an open-source tool for prospective environmental and economic life cycle assessment of vehicles. When, Where and How can battery-electric vehicles help reduce greenhouse gas emissions? *In review*
- Schmidt, O., Melchior, S., Hawkes, A., and Staffell, I. (2019). Projecting the future levelized cost of electricity storage technologies. *Joule* 3, 81–100. doi:https://doi.org/10.1016/j.joule.2018.12.008
- Staffell, I. and Pfenniger, S. (2016). Using bias-corrected reanalysis to simulate current and future wind power output. *Energy* 114, 1224–1239. doi:10.1016/j.energy.2016.08.068

- Swiss Federal Office of Energy (SFOE) (2013). *Energieperspektiven 2050 Zusammenfassung*. Tech. rep.
- [Dataset] Swiss Federal Office of Energy (SFOE) (2018). Statistik der Wasserkraftanlagen der Schweiz. Stand 1.1.2018
- [Dataset] Swiss Federal Office of Energy (SFOE) (2019a). Gesamte Erzeugung und Abgabe elektrischer Energie in der Schweiz 2018
- Swiss Federal Office of Energy (SFOE) (2019b). *Sonnendach.ch und Sonnenfassade.ch: Berechnung von Potenzialen in Gemeinden*. Tech. rep., Swiss Federal Office of Energy (SFOE)
- Swiss Federal Office of Energy (SFOE) (2019c). *Wasserkraftpotenzial der Schweiz - Abschätzung des Ausbaupotenzials der Wasserkraftnutzung im Rahmen der Energiestrategie 2050*. Tech. rep.
- Swiss Federal Office of Energy (SFOE) (2020a). *Schweizerische Elektrizitätsstatistik 2019*. Tech. rep.
- Swiss Federal Office of Energy (SFOE) (2020b). *Schweizerische Gesamtenergiestatistik 2019*. Tech. rep.
- Swiss Federal Office of Energy (SFOE) (2021). *Energieperspektiven 2050+. Kurzbericht*. Tech. rep.
- Swiss Federal Office of Energy (SFOE), Prognos AG (2012). *Die Energieperspektiven für die Schweiz bis 2050 Anhang III*. Tech. rep.
- [Dataset] Swiss Federal Office of Energy (SFOE), Prognos AG, TEP Energy GmbH, INFRAS AG (2020). Analyse des schweizerischen Energieverbrauchs 2000 - 2019 nach Verwendungszwecken
- [Dataset] Swissgrid AG (2020). Energy Statistic Switzerland 2019
- [Dataset] The Swiss Confederation, Fedlex - The publication platform for the federal law (2021). Energiegesetz (EnG) vom 30. September 2016 (Stand am 1. Januar 2021)
- Volkart, K., Mutel, C. L., and Panos, E. (2018). Integrating life cycle assessment and energy system modelling: Methodology and application to the world energy scenarios. *Sustainable Production and Consumption* 16, 121–133. doi:10.1016/j.spc.2018.07.001
- Wernet, G., Bauer, C., Steubing, B., Reinhard, J., Moreno-Ruiz, E., and Weidmann, B. (2016). The ecoinvent database version 3 (part I): overview and methodology. *The International Journal of Life Cycle Assessment* 21, 1218–1230. doi:10.1007/s11367-016-1087-8